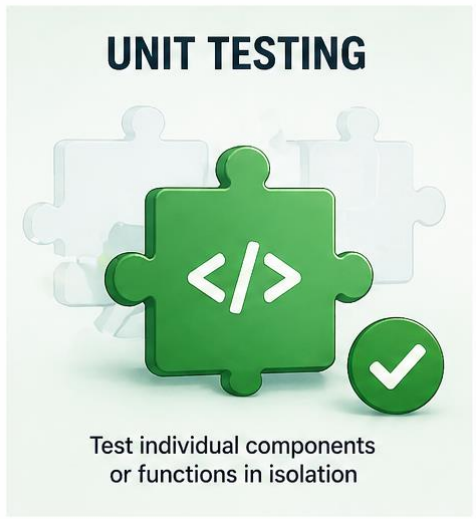
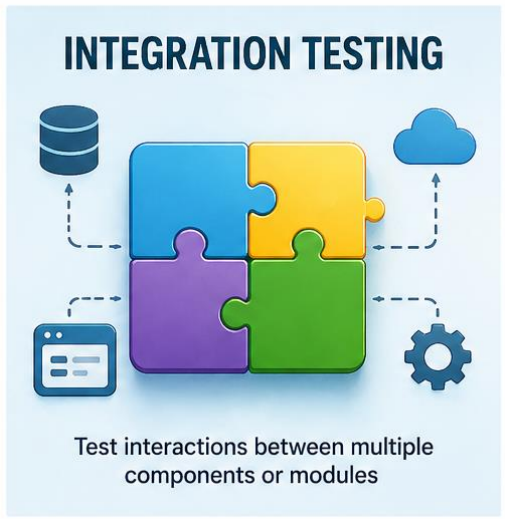
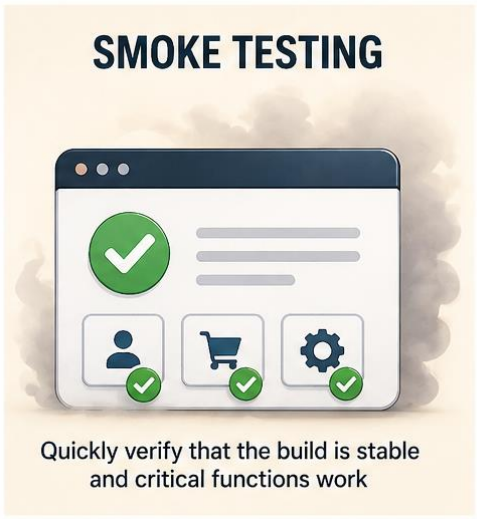
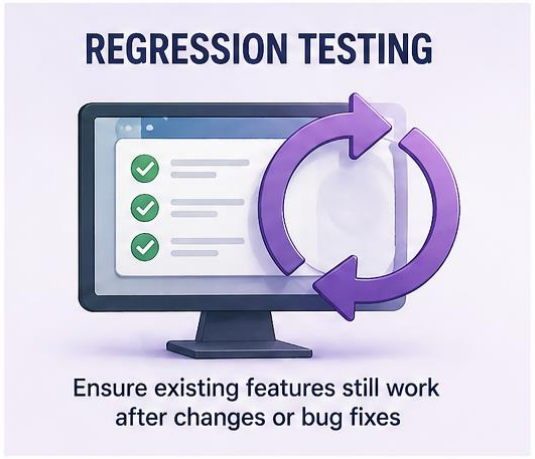
Testing

Testing


UNIT TESTING
Test individual components
or functions in isolation


INTEGRATION TESTING
Test interactions between multiple
components or modules


SMOKE TESTING
Quickly verify that the build is stable
and critical functions work

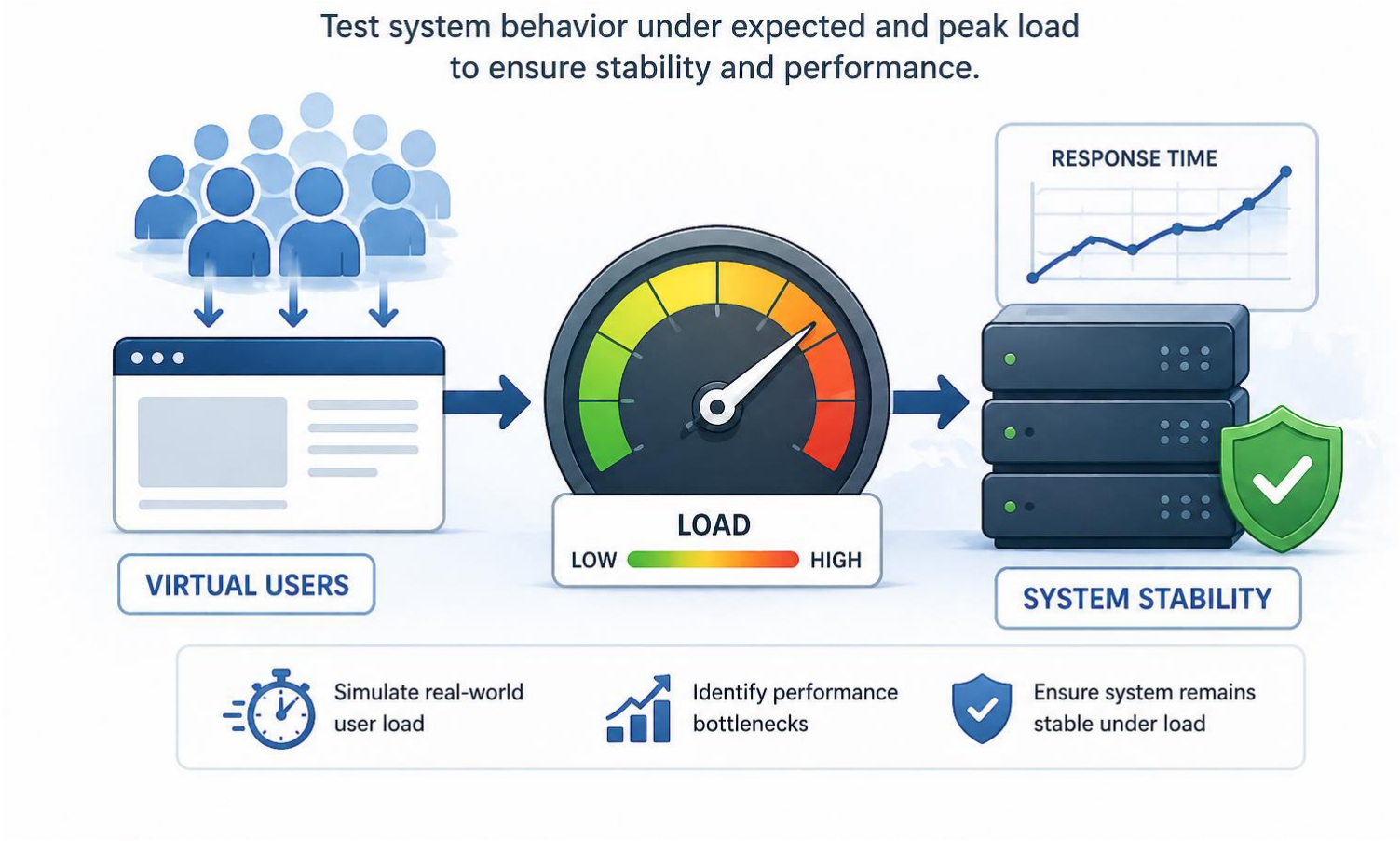

REGRESSION TESTING
Ensure existing features still work
after changes or bug fixes


PERFORMANCE TESTING
Evaluate speed, stability, scalability
and resource usage under load

Testing

1. Unit Testing
   - Unit testing focuses on verifying **individual components** or functions of the software in isolation. It ensures that each unit performs as expected independently. Typically, developers write and run these tests during the **development** phase
2. Integration Testing
   - Integration testing assesses the **interaction between different modules** or services within the application. It ensures that combined components function together correctly, identifying issues in interfaces and data flow between modules
3. Smoke Testing
   - Smoke testing involves a preliminary check to ensure the **basic functionalities** of the application work correctly. It's often referred to as a "sanity check" before proceeding to more rigorous testing.
4. Regression Testing
   - Regression testing ensures that recent code changes **haven't adversely affected existing functionalities**. It involves re-running previous test cases to confirm that the software continues to perform as expected.
5. Performance Testing
   - Performance testing evaluates the software's responsiveness, stability, and scalability under various load conditions. It includes:
     - **Load Testing**: Assesses the system’s behaviour under expected user loads.
     - **Stress Testing**: Determines the system's robustness by testing beyond normal operational capacity. To Find the system’s *breaking point.*
     - **Endurance Testing**: Checks the system's stability over an extended period of sustained load.

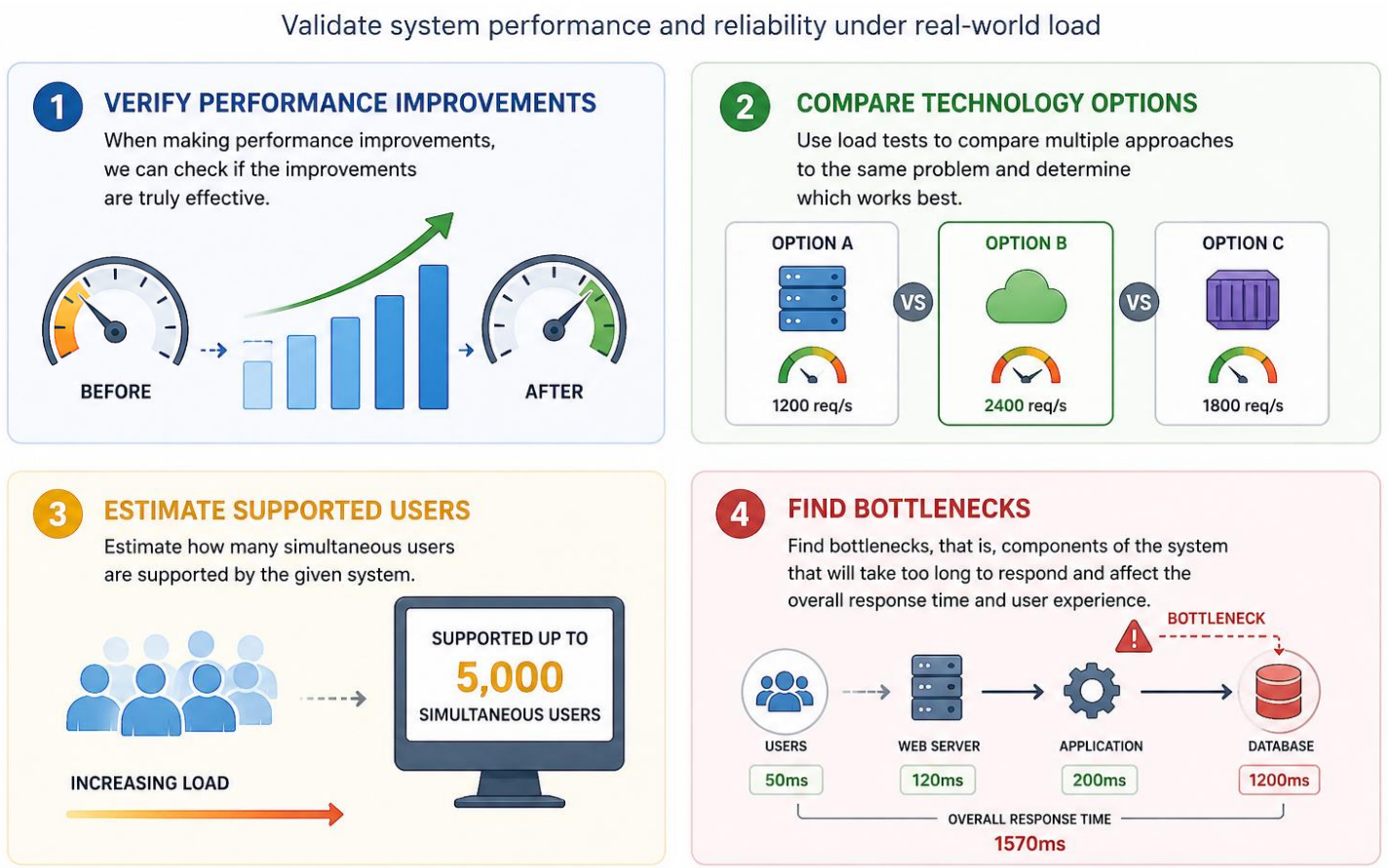
Load Testing



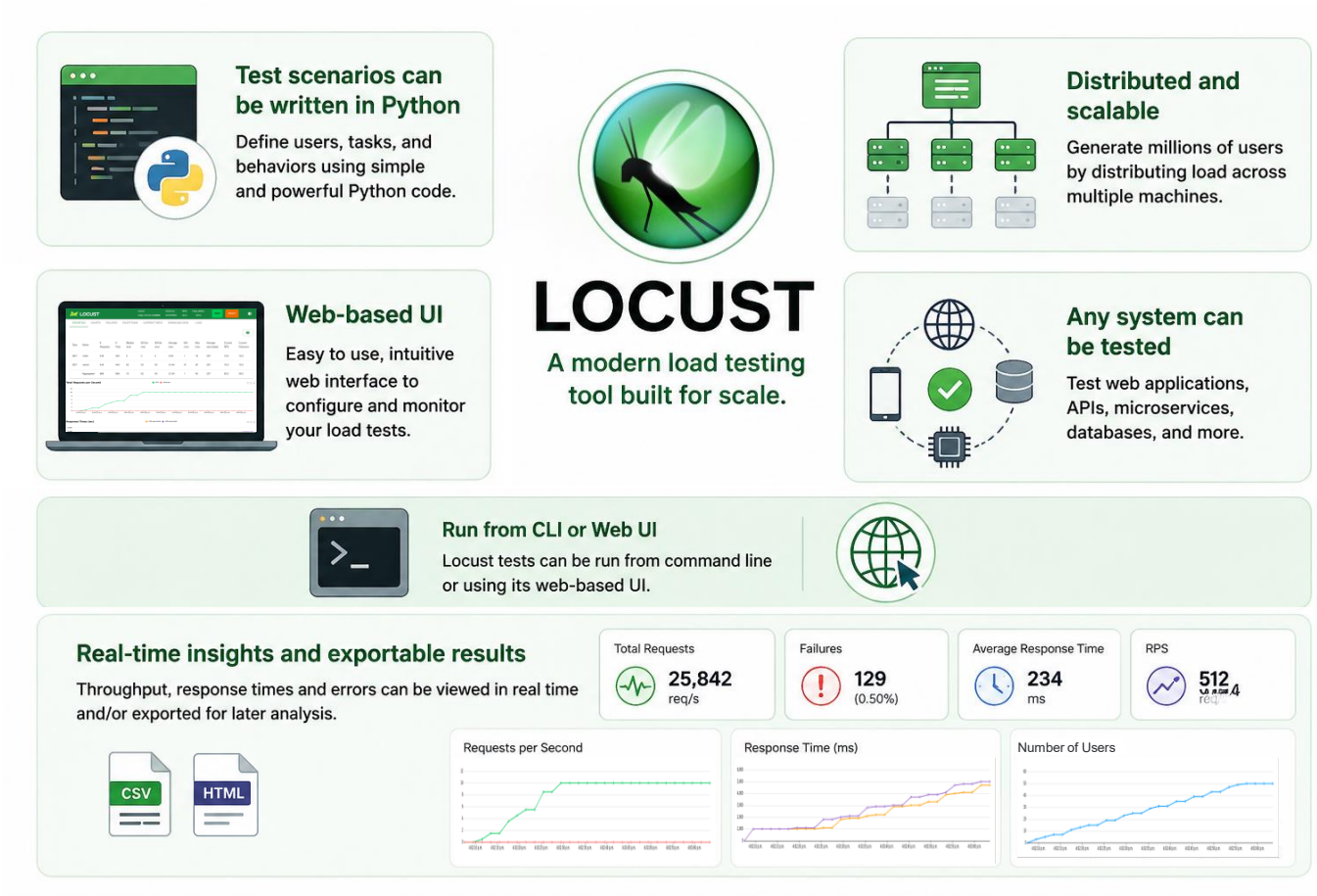
As an AI Engineer, how can I benefit from load testing?

Load Testing

When should I use load tests?



Load Testing – Locust



Load Testing – Locust

Behavior 1

User

User

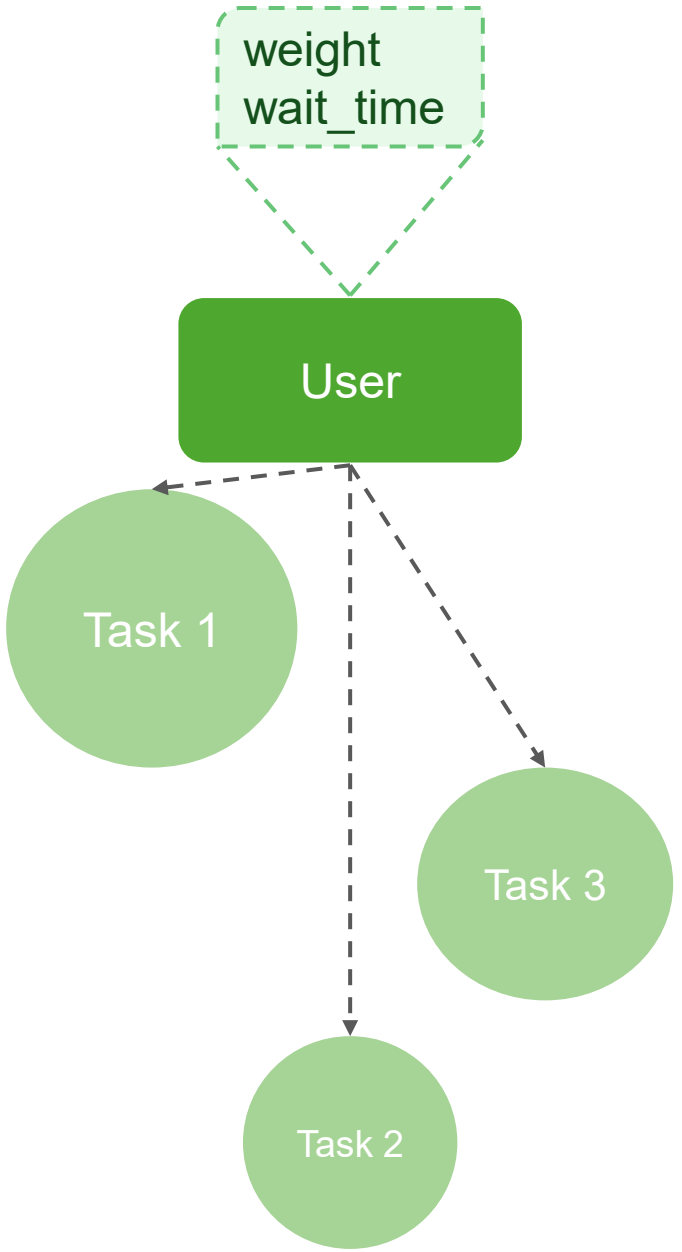
.....

Behavior 2

User

User

.....



Load Testing – Locust 

A user class represents one type of user/scenario for your system. When you do a test run you specify the number of concurrent users you want to simulate and Locust will create an instance per user. Each of these users will start running within their own green gevent thread.

User classes:

- HttpUser
  - Gives each user a client attribute
- FastHttpUser
  - Uses geventhttpclient instead of python-requests as its underlying client.
  - It uses considerably less CPU on the load generator, and should work as a simple drop-in-replacement in most cases.

Load Testing – Locust 

The self.client attribute makes it possible to make HTTP calls that will be logged by Locust
It contains methods for all HTTP methods: **get**, **post**, **put**, …

```python
from locust import HttpUser

class MyUser(HttpUser):


    @task
    def index(self):
        self.client.get("/")

    @task
    def about(self):
        self.client.get("/about/")
```

Load Testing – Locust 

You can add any attributes you like to the user classes/instances, but there are some that have special meaning to Locust:

- **weight**
  - If you wish to simulate more users of a certain type than another, you can set a weight attribute on those classes.
  - If more than one user class exists in the file, and no user classes are specified on the command line, Locust will spawn *an equal number* of each of the user classes.

- **fixed_count**
  - If you set the fixed_count attribute, the weight attribute will be ignored and only that exact number users will be spawned.
  - These users are spawned before any regular, weighted ones.

```
class WebUser(User):
    weight = 3
    ...

class MobileUser(User):
    weight = 1
    ...
```

```
class AdminUser(User):
    wait_time = constant(600)
    fixed_count = 1

    @task
    def restart_app(self):
        ...

class WebUser(User):
    ...
```

Load Testing – Locust 

You can add any attributes you like to the user classes/instances, but there are some that have special meaning to Locust:

- **wait_time**
  - If no *wait_time* is specified, the next task will be executed as soon as one finishes.
  - *constant:* for a fixed amount of time
  - *between:* for a random time between a min and max value
  - *constant_pacing:* for an adaptive time that ensures the task runs (at most) once every X seconds (it is the mathematical inverse of constant_throughput).
  - *constant_throughput:* for an adaptive time that ensures the task runs (at most) X times per second.

```python
from locust import HttpUser, task, between

class MyUser(HttpUser):
    wait_time = between(5, 15)

    @task(4)
    def index(self):
        self.client.get("/")

    @task(1)
    def about(self):
        self.client.get("/about/")
```

```python
class AdminUser(User):
    wait_time = constant(600)
    fixed_count = 1

    @task
    def restart_app(self):
        ...

class WebUser(User):
    ...
```

```python
class MyUser(User):
    wait_time = constant_pacing(10)
    @task
    def my_task(self):
        time.sleep(random.random())
```

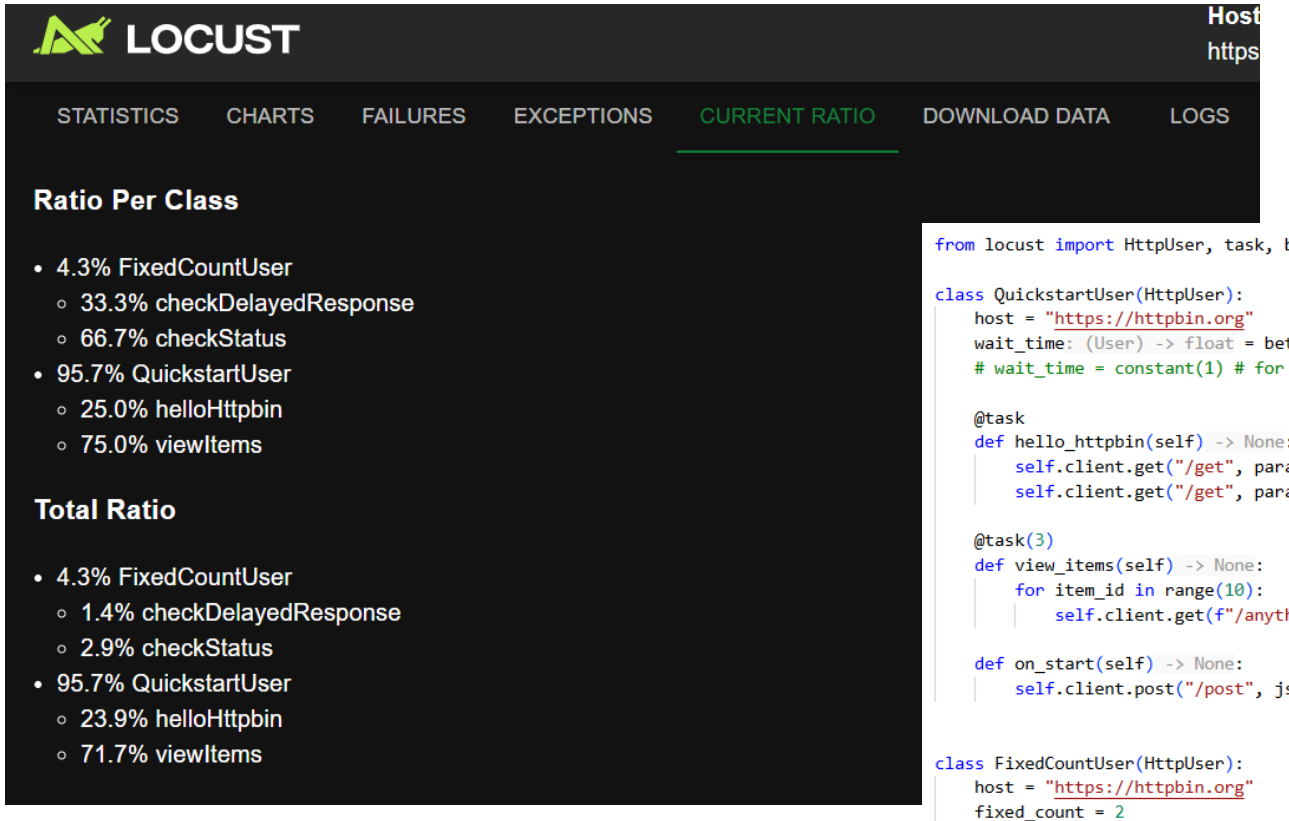
Let’s Code IT

Load Testing – Locust 

Installation

```
> pip install locust
```

Load Testing – Locust




```python
from locust import HttpUser, task, between, constant

class QuickstartUser(HttpUser):
    host = "https://httpbin.org"
    wait_time: (User) -> float = between(1, 5) # for a random time between a min and max value in seconds
    # wait_time = constant(1) # for a fixed amount of time

    @task
    def hello_httpbin(self) -> None:
        self.client.get("/get", params={"source": "quickstart", "route": "hello"})
        self.client.get("/get", params={"source": "quickstart", "route": "world"})

    @task(3)
    def view_items(self) -> None:
        for item_id in range(10):
            self.client.get(f"/anything/item/{item_id}", name="/anything")

    def on_start(self) -> None:
        self.client.post("/post", json={"username":"foo", "password":"bar"})


class FixedCountUser(HttpUser):
    host = "https://httpbin.org"
    fixed_count = 2
    wait_time: (User) -> float = constant(1)

    @task(2)
    def check_status(self) -> None:
        self.client.get("/status/200", name="/status")

    @task
    def check_delayed_response(self) -> None:
        self.client.get("/delay/1", name="/delay")
```

Use Case

Load Testing – Locust

Locust Test Report

During: 11/4/2021, 9:03:39 AM - 11/4/2021, 9:11:17 AM

Target Host: https://django-todo-list-grjv6.ondigitalocean.app

Script: locustfile.py

Download the Report

Request Statistics

Method	Name	# Requests	# Fails	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	RPS	Failures/s
GET	/tasks/	21512	3	1980	36	13291	929	47.0	0.0
POST	/tasks/	4360	1	2002	14	13276	79	9.5	0.0
	Aggregated	25872	4	1984	14	13291	786	56.5	0.0

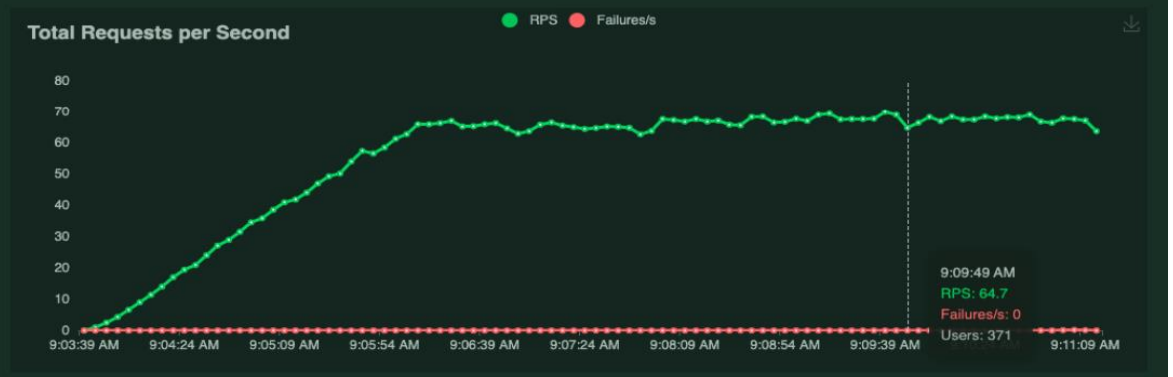
Response Time Statistics

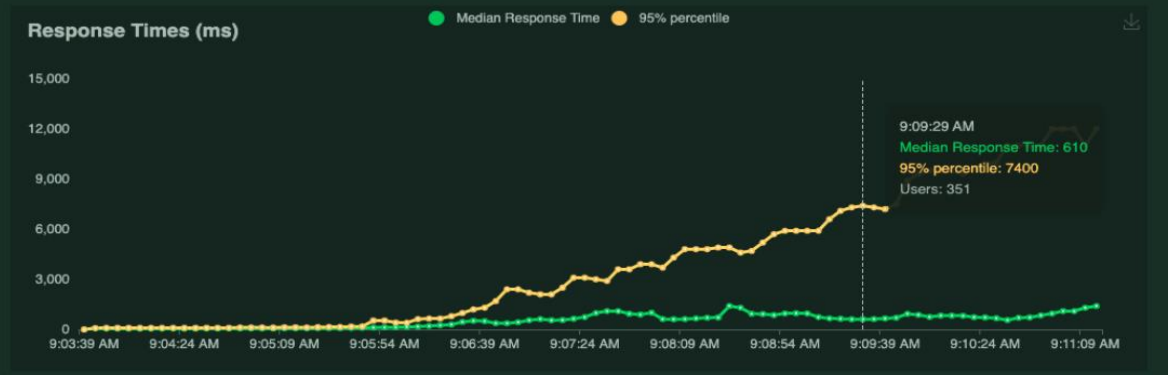
Method	Name	50%ile (ms)	60%ile (ms)	70%ile (ms)	80%ile (ms)	90%ile (ms)	95%ile (ms)	99%ile (ms)	100%ile (ms)
GET	/tasks/	550	740	1200	3600	7100	9800	12000	13000
POST	/tasks/	540	740	1200	3700	7100	9700	12000	13000
	Aggregated	550	740	1200	3600	7100	9800	12000	13000

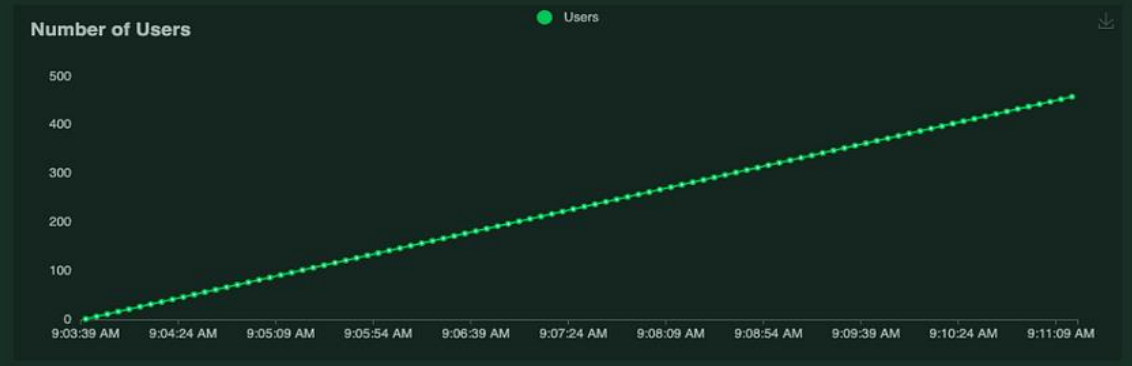
Failures Statistics

Method	Name	Error	Occurrences
GET	/tasks/	504 Server Error: Gateway Time-out for url: https://django-todo-list-grjv6.ondigitalocean.app/tasks/	3
POST	/tasks/	504 Server Error: Gateway Time-out for url: https://django-todo-list-grjv6.ondigitalocean.app/tasks/	1

Charts







Tasks

Ratio per User class

- 100.0% WebsiteUser
  - 83.3% index
  - 16.7% create

Total ratio

- 100.0% WebsiteUser
  - 83.3% index

Load Testing – Locust 

- Although the user number is increasing the RPS is still and there are no failures. Where does all the rest of the users go??

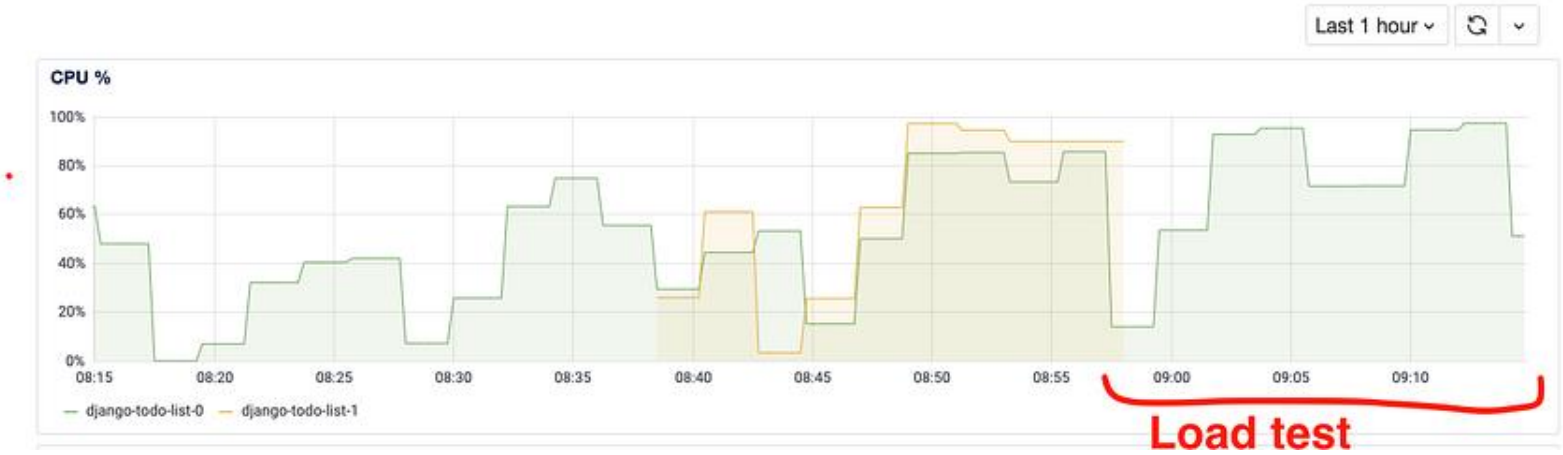
Load Testing – Locust 

Results and conclusions

- Until around 160 simultaneous users, we were having around 60 requests per second and the response time was very acceptable for this context (95% of requests finished in less than 400ms)
- After passing 160 simultaneous users, note the number of requests per second remains the same (60rps), but the response time increased a lot. This way, each user started having slower responses.
- In summary, with the available computing resources, we could serve around 160 simultaneous users with 60rps and a response time lower than 400ms. To increase these numbers, we would need more computing resources or make performance adjustments in the code.

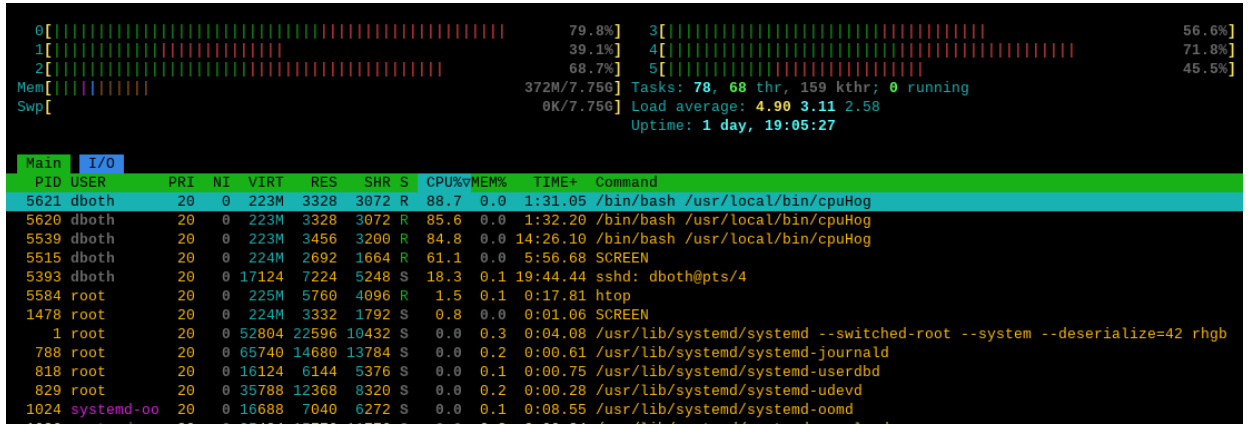
Load Testing – Locust

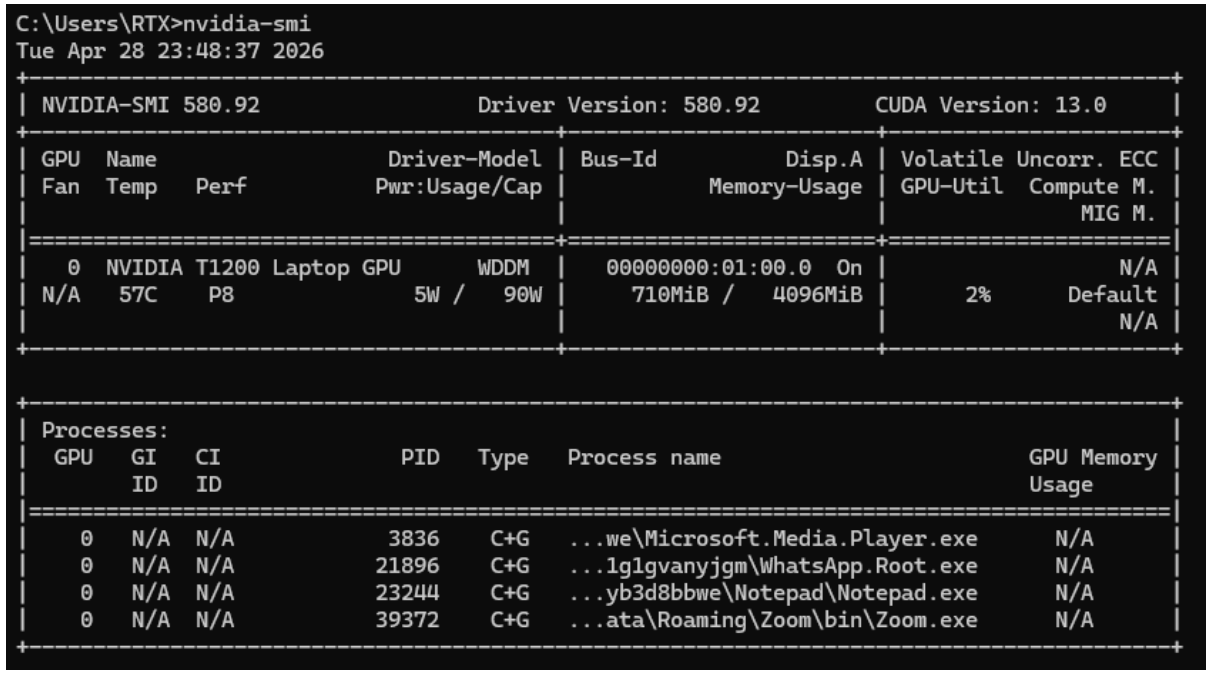
- In our application layer, we see the CPU usage growing until it reached 100%. In a way, it's a good sign that CPU usage is ramping up. This means our application server is well configured and it is not leaving idle resources.
- On the other hand, since our CPU usage is getting to 100%, this also means we need more computing resources.



Load Testing – Locust

Commands to use: Htop & nvidia-smi

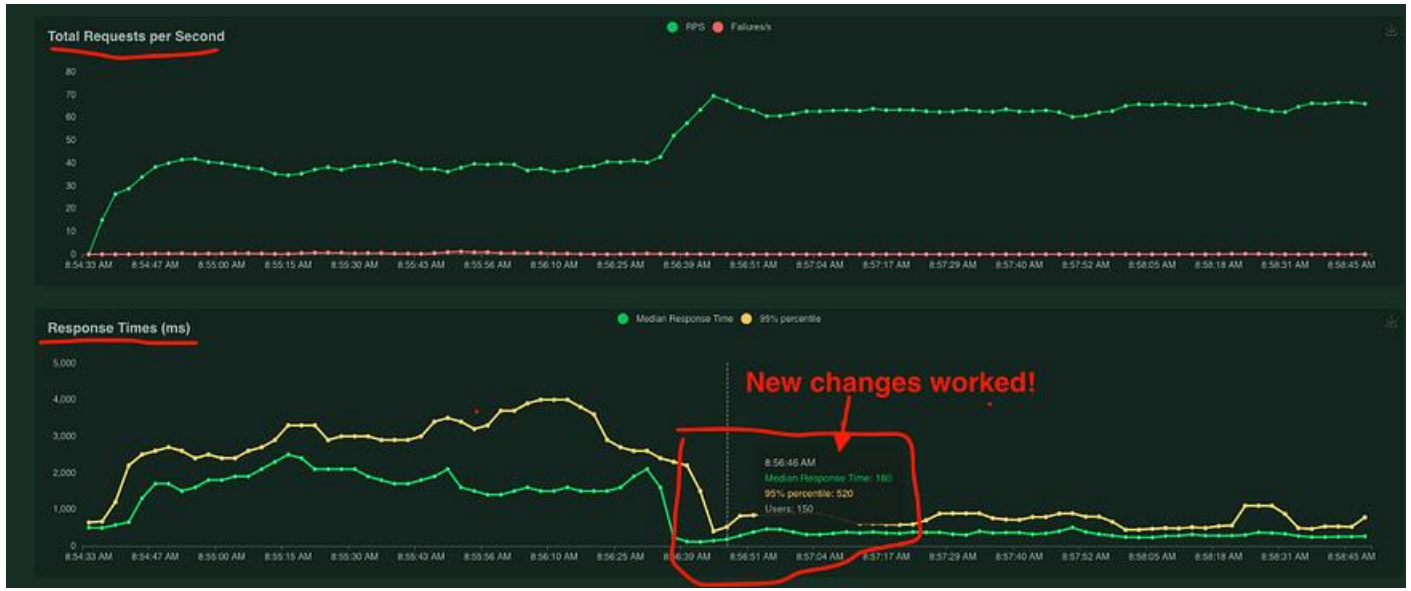




Load Testing – Locust

One step at a time

- When performing load tests and improving performance, it is important to make **small changes, one at a time**. This way we can verify how much a given change affected the results. In the screenshot below, we can see the exact point where I changed a configuration and the response time improved a lot:
- Combining small changes with monitoring (results and computing resources), we can understand how our system behaves under stress and adjust accordingly.



LAB

Load Testing – Lab

- What you'll do is to load test the local deployed container of churn model of the previous lab:
  - Test your deployment through locust-python script
  - Observe the charts and experiment with different parameter values e.g. number of users
  - Submit your answer to those questions:
    - What's the peak RPS?
    - What other insights did you observe from the graph?

Additional Resources

- Load Testing Using Locust - Use Case
- Locust Documentation